# Empirical Applications (1): World Trade Center

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## Basics: Modeling Phone Calls in an Emergency Context

### Preliminarities

The following section aims to load the necessary packages and import the data. Furthermore, we modify the data to facilitate their use within `eventnet`.

Specifically: - We include rows that list the actors in the system. This allows us to construct the risk set. - We provide the exogenous information in a format that can be read as actor-specific attributes by `eventnet`.

```
load(file="input/WTC.RData")
```

```
data <- WTCPoliceCalls[,c("source", "recipient", "number")]
colnames(data) <- c("s1", "r1", "time")
data$type <- "call"
```

```
num.actors <- length(WTCPoliceIsICR)
actors <- 1:num.actors
```

```
before.min.time <- min(data$time) - 1
dummy.events <- data.frame(actors, actors,
                           rep(before.min.time,num.actors),
                           rep("add.actor",num.actors))
colnames(dummy.events) <- colnames(data)
head(dummy.events)
```

```
events <- rbind(dummy.events, data)
```

```
ICR <- which(WTCPoliceIsICR)
num.ICR <- length(ICR)
dummy.events <- data.frame(ICR, ICR,
                           rep(before.min.time,num.ICR),
                           rep("is.ICR",num.ICR))
colnames(dummy.events) <- colnames(events)
events <- rbind(dummy.events, events)
```

```
events <- events[order(events$time),]
write.csv(events, file="input/WTC_data.csv", row.names = F)
```

The dataset `WTC_data.csv` is then processed in `eventnet`, and the corresponding configuration file can be found at `eventnet/configuration_WTC.txt`.

## Model fitting

```
data_original <- read.csv("input/WTC_data_FULL.csv")
data_full <- data_original[,setdiff(colnames(data_original),
                                    c("TYPE",
                                      "EVENT_INTERVAL_ID", "EVENT",
                                      "INTEGER_TIME", "TIME_POINT",
                                      "TIME_UNIT", "num.actors"))]
rm(data_original)
head(data_full)
```

```
##   IS_OBSERVED SOURCE TARGET TIME EVENT_INTERVAL sender_activity
## 1           1     32     16    2              2               0
## 2           0     22     23    2              2               0
## 3           0     22     24    2              2               0
## 4           0     22     25    2              2               0
## 5           0     22     26    2              2               0
## 6           0     22     27    2              2               0
##   receiver_popularity repetition reciprocity is.ICRs is.ICRr closure
## 1                   0          0           1       1       0       0
## 2                   0          0           0       0       0       0
## 3                   0          0           0       0       0       0
## 4                   0          0           0       0       0       0
## 5                   0          0           0       0       0       0
## 6                   0          0           0       0       0       0
```

```
clogit_fit_full <- clogit(IS_OBSERVED ~
                          + sender_activity
                          + receiver_popularity
                          + repetition
                          + reciprocity
                          + closure
                          + is.ICRs
                          + is.ICRr
                          + strata(EVENT_INTERVAL)
                          , data = data_full)
s_clogit_fit_full <- summary(clogit_fit_full)
```

### Results

```
s_clogit_fit_full
```

```
## Call:
## coxph(formula = Surv(rep(1, 639360L), IS_OBSERVED) ~ +sender_activity +
##     receiver_popularity + repetition + reciprocity + closure +
##     is.ICRs + is.ICRr + strata(EVENT_INTERVAL), data = data_full,
##     method = "exact")
##
##   n= 639360, number of events= 480
##
##                           coef  exp(coef)   se(coef)      z Pr(>|z|)
## sender_activity      0.0357692  1.0364166  0.0027732 12.898  < 2e-16 ***
## receiver_popularity  0.0295470  1.0299878  0.0021782 13.565  < 2e-16 ***
## repetition          -0.2394916  0.7870279  0.0300588 -7.967 1.62e-15 ***
## reciprocity          0.2550572  1.2905354  0.0293581  8.688  < 2e-16 ***
```

```
## closure              0.0023523  1.0023551  0.0005121  4.594 4.35e-06 ***
## is.ICRs              0.5336643  1.7051691  0.1735244  3.075   0.0021 **
## is.ICRr              1.2604475  3.5269995  0.1573502  8.010 1.14e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                     exp(coef) exp(-coef) lower .95 upper .95
## sender_activity         1.036     0.9649     1.031    1.0421
## receiver_popularity     1.030     0.9709     1.026    1.0344
## repetition              0.787     1.2706     0.742    0.8348
## reciprocity             1.291     0.7749     1.218    1.3670
## closure                 1.002     0.9977     1.001    1.0034
## is.ICRs                 1.705     0.5865     1.214    2.3959
## is.ICRr                 3.527     0.2835     2.591    4.8011
##
## Concordance= 0.941  (se = 0.006 )
## Likelihood ratio test= 1985  on 7 df,   p=<2e-16
## Wald test            = 1534  on 7 df,   p=<2e-16
## Score (logrank) test = 26251  on 7 df,   p=<2e-16
```

```r
names <- c("Sender Activity","Receiver Popularity","Repetition",
           "Reciprocity", "Closure", "Sender is ICR", "Receiver is ICR")

summary <- as.data.frame(s_clogit_fit_full$coefficients[,c(1:3)])
summary <- round(summary, 3)
summary <- cbind(rownames(s_clogit_fit_full$coefficients), summary)
colnames(summary) <- c("Explanatory Variable",
                       "Estimate",
                       "Exp(Estimate)",
                       "S.E.(Estimate)")
rownames(summary) <- NULL
```

```r
ft <- flextable(summary) %>%
  set_table_properties(layout = "fixed", width = 1) %>%
  width(j = c(1), width = 2) %>%
  fontsize(size = 10, part = "all")
save_as_image(ft, path="pictures/empirical/WTC/summary_WTC.png")
```

```
## [1] "pictures/empirical/WTC/summary_WTC.png"
```

### Case-m-control sampling

```r
ensure_non_events <- function(data_sampled, data_full, m = 1, seed = 1234) {

  set.seed(seed)

  # Separate events and non-events from sampled data
  events_sampled <- data_sampled %>% filter(IS_OBSERVED == 1)
  non_events_sampled <- data_sampled %>% filter(IS_OBSERVED == 0)

  # Group events by TIME and EVENT_INTERVAL
  events_grouped <- events_sampled %>%
    distinct(TIME, EVENT_INTERVAL)
```

```r
  # For each event group, check number of available non-events
  additions <- events_grouped %>%
    rowwise() %>%
    do({
      t <- .$TIME
      ei <- .$EVENT_INTERVAL

      # Existing non-events for this (TIME, EVENT_INTERVAL)
      current_non_events <- non_events_sampled %>%
        filter(TIME == t, EVENT_INTERVAL == ei)

      n_missing <- m - nrow(current_non_events)

      if (n_missing > 0) {
        # Get candidates from full data
        candidates <- data_full %>%
          filter(IS_OBSERVED == 0, TIME == t, EVENT_INTERVAL == ei)

        # Sample min(n_missing, number of available candidates)
        n_to_sample <- min(n_missing, nrow(candidates))

        if (n_to_sample > 0) {
          sampled <- candidates %>% sample_n(n_to_sample)
          sampled
        } else {
          tibble()  # no candidates available
        }
      } else {
        tibble()  # no missing non-events
      }
    }) %>%
    ungroup()

  # Combine original sampled data with additions
  data_fixed <- bind_rows(data_sampled, additions)

  return(data_fixed)
}
```

```r
sum(data_full$IS_OBSERVE==1)
```

```
## [1] 480
```

```r
nrow(data_full)/480
```

```
## [1] 1332
```

```r
37*37 - 37
```

```
## [1] 1332
```

### m = 1

```r
data_original <- read.csv("input/WTC_data_NCC_1.csv")
data_sampled_1 <- data_original[,setdiff(colnames(data_original),
                                         c("TYPE", "EVENT_INTERVAL_ID", "EVENT",
```

```r
                                        "INTEGER_TIME", "TIME_POINT",
                                        "TIME_UNIT", "num.actors"))]
```

```r
sum(data_sampled_1$IS_OBSERVED==0)
```

```
## [1] 464
```

```r
sum(data_sampled_1$IS_OBSERVED==1)
```

```
## [1] 480
```

```r
data_sampled_1 <- ensure_non_events(data_sampled_1, data_full,
                                    m = 1, seed = 1234)
```

```r
sum(data_sampled_1$IS_OBSERVED==0)
```

```
## [1] 480
```

```r
sum(data_sampled_1$IS_OBSERVED==1)
```

```
## [1] 480
```

```r
rm(data_original)
```

```r
clogit_fit_m1 <- clogit(IS_OBSERVED ~
                          + sender_activity
                        + receiver_popularity
                        + repetition
                        + reciprocity
                        + closure
                        + is.ICRs
                        + is.ICRr
                        + strata(EVENT_INTERVAL)
                        , data = data_sampled_1)
s_clogit_fit_m1 <- summary(clogit_fit_m1)
s_clogit_fit_m1
```

```
## Call:
## coxph(formula = Surv(rep(1, 960L), IS_OBSERVED) ~ +sender_activity +
##     receiver_popularity + repetition + reciprocity + closure +
##     is.ICRs + is.ICRr + strata(EVENT_INTERVAL), data = data_sampled_1,
##     method = "exact")
##
##   n= 960, number of events= 480
##
##                          coef exp(coef)  se(coef)      z Pr(>|z|)
## sender_activity      0.046616  1.047720  0.014714  3.168  0.00153 **
## receiver_popularity  0.039866  1.040671  0.012915  3.087  0.00202 **
## repetition          -0.168936  0.844563  0.289684 -0.583  0.55978
## reciprocity          1.027692  2.794610  0.317763  3.234  0.00122 **
## closure             -0.001793  0.998209  0.004361 -0.411  0.68101
## is.ICRs              0.297913  1.347045  0.517580  0.576  0.56489
## is.ICRr              0.901078  2.462256  0.511945  1.760  0.07839 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                     exp(coef) exp(-coef) lower .95 upper .95
```

```
## sender_activity        1.0477     0.9545    1.0179     1.078
## receiver_popularity    1.0407     0.9609    1.0147     1.067
## repetition             0.8446     1.1840    0.4787     1.490
## reciprocity            2.7946     0.3578    1.4991     5.210
## closure                0.9982     1.0018    0.9897     1.007
## is.ICRs                1.3470     0.7424    0.4884     3.715
## is.ICRr                2.4623     0.4061    0.9027     6.716
##
## Concordance= 0.968  (se = 0.011 )
## Likelihood ratio test= 548.2  on 7 df,   p=<2e-16
## Wald test            = 55.44  on 7 df,   p=1e-09
## Score (logrank) test = 325  on 7 df,   p=<2e-16
```

### $\mathbf{m = 5}$

```r
data_original <- read.csv("input/WTC_data_NCC_5.csv")
data_sampled_5 <- data_original[,setdiff(colnames(data_original),
                                         c("TYPE", "EVENT_INTERVAL_ID", "EVENT",
                                           "INTEGER_TIME", "TIME_POINT",
                                           "TIME_UNIT", "num.actors"))]
rm(data_original)
```

```r
sum(data_sampled_5$IS_OBSERVED==0)
```

```
## [1] 2324
```

```r
sum(data_sampled_5$IS_OBSERVED==1) * 5
```

```
## [1] 2400
```

```r
data_sampled_5 <- ensure_non_events(data_sampled_5, data_full,
                                    m = 5, seed = 1)
```

```r
sum(data_sampled_5$IS_OBSERVED==0)
```

```
## [1] 2400
```

```r
sum(data_sampled_5$IS_OBSERVED==1) * 5
```

```
## [1] 2400
```

```r
rm(data_original)
```

```
## Warning in rm(data_original): object 'data_original' not found
```

```r
clogit_fit_m5 <- clogit(IS_OBSERVED ~
                          + sender_activity
                        + receiver_popularity
                        + repetition
                        + reciprocity
                        + closure
                        + is.ICRs
                        + is.ICRr
                        + strata(EVENT_INTERVAL)
                        , data = data_sampled_5)
s_clogit_fit_m5 <- summary(clogit_fit_m5)
s_clogit_fit_m5
```

```
## Call:
## coxph(formula = Surv(rep(1, 2880L), IS_OBSERVED) ~ +sender_activity +
##     receiver_popularity + repetition + reciprocity + closure +
##     is.ICRs + is.ICRr + strata(EVENT_INTERVAL), data = data_sampled_5,
##     method = "exact")
##
##   n= 2880, number of events= 480
##
##                          coef exp(coef)  se(coef)      z Pr(>|z|)
## sender_activity      0.019667  1.019861  0.005044  3.899 9.66e-05 ***
## receiver_popularity  0.024305  1.024603  0.004403  5.520 3.39e-08 ***
## repetition          -0.286606  0.750807  0.156671 -1.829   0.0673 .
## reciprocity          1.309855  3.705636  0.183112  7.153 8.47e-13 ***
## closure              0.001505  1.001506  0.001469  1.025   0.3054
## is.ICRs              0.168939  1.184048  0.335471  0.504   0.6146
## is.ICRr              1.364988  3.915677  0.294561  4.634 3.59e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                     exp(coef) exp(-coef) lower .95 upper .95
## sender_activity        1.0199     0.9805    1.0098     1.030
## receiver_popularity    1.0246     0.9760    1.0158     1.033
## repetition             0.7508     1.3319    0.5523     1.021
## reciprocity            3.7056     0.2699    2.5882     5.306
## closure                1.0015     0.9985    0.9986     1.004
## is.ICRs                1.1840     0.8446    0.6135     2.285
## is.ICRr                3.9157     0.2554    2.1982     6.975
##
## Concordance= 0.967  (se = 0.007 )
## Likelihood ratio test= 1301  on 7 df,   p=<2e-16
## Wald test            = 210.3  on 7 df,   p=<2e-16
## Score (logrank) test = 1253  on 7 df,   p=<2e-16
```

### m = 20

```r
data_original <- read.csv("input/WTC_data_NCC_20.csv")
data_sampled_20 <- data_original[,setdiff(colnames(data_original),
                                          c("TYPE", "EVENT_INTERVAL_ID", "EVENT",
                                            "INTEGER_TIME", "TIME_POINT",
                                            "TIME_UNIT", "num.actors"))]
```

```r
sum(data_sampled_20$IS_OBSERVED==0)
```

```
## [1] 9338
```

```r
sum(data_sampled_20$IS_OBSERVED==1) * 20
```

```
## [1] 9600
```

```r
data_sampled_20 <- ensure_non_events(data_sampled_5, data_full,
                                     m = 20, seed = 1234)
```

```r
sum(data_sampled_20$IS_OBSERVED==0)
```

```
## [1] 9600
```

```r
sum(data_sampled_20$IS_OBSERVED==1) * 20
```

```
## [1] 9600
```

```r
clogit_fit_m20 <- clogit(IS_OBSERVED ~
                          + sender_activity
                        + receiver_popularity
                        + repetition
                        + reciprocity
                        + closure
                        + is.ICRs
                        + is.ICRr
                        + strata(EVENT_INTERVAL)
                        , data = data_sampled_20)
s_clogit_fit_m20 <- summary(clogit_fit_m20)
s_clogit_fit_m20
```

```
## Call:
## coxph(formula = Surv(rep(1, 10080L), IS_OBSERVED) ~ +sender_activity +
##     receiver_popularity + repetition + reciprocity + closure +
##     is.ICRs + is.ICRr + strata(EVENT_INTERVAL), data = data_sampled_20,
##     method = "exact")
##
##   n= 10080, number of events= 480
##
##                          coef  exp(coef)   se(coef)      z Pr(>|z|)
## sender_activity      0.0246743  1.0249812  0.0030927  7.978 1.49e-15 ***
## receiver_popularity  0.0227409  1.0230014  0.0024771  9.180  < 2e-16 ***
## repetition          -0.3220257  0.7246796  0.0833188 -3.865 0.000111 ***
## reciprocity          0.5353488  1.7080439  0.0837593  6.392 1.64e-10 ***
## closure              0.0033877  1.0033935  0.0006912  4.901 9.53e-07 ***
## is.ICRs              0.3552965  1.4266036  0.2217912  1.602 0.109168
## is.ICRr              1.2957123  3.6535975  0.1933234  6.702 2.05e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                     exp(coef) exp(-coef) lower .95 upper .95
## sender_activity        1.0250     0.9756    1.0188    1.0312
## receiver_popularity    1.0230     0.9775    1.0180    1.0280
## repetition             0.7247     1.3799    0.6155    0.8532
## reciprocity            1.7080     0.5855    1.4494    2.0128
## closure                1.0034     0.9966    1.0020    1.0048
## is.ICRs                1.4266     0.7010    0.9237    2.2034
## is.ICRr                3.6536     0.2737    2.5013    5.3368
##
## Concordance= 0.949  (se = 0.006 )
## Likelihood ratio test= 1642  on 7 df,   p=<2e-16
## Wall test            = 563.4  on 7 df,   p=<2e-16
## Score (logrank) test = 3511  on 7 df,   p=<2e-16
```

### Results

```r
# Extract coefficients
coef_full <- coefficients(clogit_fit_full)
```

```r
coef_m20  <- coefficients(clogit_fit_m20)
coef_m5   <- coefficients(clogit_fit_m5)
coef_m1   <- coefficients(clogit_fit_m1)

# Extract standard errors
se_full <- s_clogit_fit_full$coefficients[,3]
se_m20  <- s_clogit_fit_m20$coefficients[,3]
se_m5   <- s_clogit_fit_m5$coefficients[,3]
se_m1   <- s_clogit_fit_m1$coefficients[,3]

# Variable names
variables <- names(coef_full)

# Create table with formatted coefficient (SE)
results_table <- data.frame(
  Variable = variables,
  `Full Model` = sprintf("%.3f (%.3f)", coef_full, se_full),
  `m20 Model`  = sprintf("%.3f (%.3f)", coef_m20, se_m20),
  `m5 Model`   = sprintf("%.3f (%.3f)", coef_m5, se_m5),
  `m1 Model`   = sprintf("%.3f (%.3f)", coef_m1, se_m1),
  stringsAsFactors = FALSE
)
```

```r
latex_table <- xtable(results_table,
                      caption = "Coefficients and Standard Errors from Conditional Logistic Regression M
                      label = "tab:clogit_results",
                      align = c("l", "l", "c", "c", "c", "c"))
print(latex_table, include.rownames = FALSE,
      sanitize.text.function = identity)
```

```
## % latex table generated in R 4.2.3 by xtable 1.8-4 package
## % Mon Apr  6 19:59:07 2026
## \begin{table}[ht]
## \centering
## \begin{tabular}{lcccc}
##   \hline
## Variable & Full.Model & m20.Model & m5.Model & m1.Model \\
##   \hline
## sender_activity & 0.036 (0.003) & 0.025 (0.003) & 0.020 (0.005) & 0.047 (0.015) \\
##   receiver_popularity & 0.030 (0.002) & 0.023 (0.002) & 0.024 (0.004) & 0.040 (0.013) \\
##   repetition & -0.239 (0.030) & -0.322 (0.083) & -0.287 (0.157) & -0.169 (0.290) \\
##   reciprocity & 0.255 (0.029) & 0.535 (0.084) & 1.310 (0.183) & 1.028 (0.318) \\
##   closure & 0.002 (0.001) & 0.003 (0.001) & 0.002 (0.001) & -0.002 (0.004) \\
##   is.ICRs & 0.534 (0.174) & 0.355 (0.222) & 0.169 (0.335) & 0.298 (0.518) \\
##   is.ICRr & 1.260 (0.157) & 1.296 (0.193) & 1.365 (0.295) & 0.901 (0.512) \\
##    \hline
## \end{tabular}
## \caption{Coefficients and Standard Errors from Conditional Logistic Regression Models}
## \label{tab:clogit_results}
## \end{table}
```